

Extended Reality System for Robotic Learning from Human Demonstration

Isaac Ngui*

University of Illinois Urbana-Champaign

André Corrêa Santos[§]

Insper

Courtney McBeth[†]

University of Illinois Urbana-Champaign

Luciano Soares[¶]

Insper

Grace He[‡]

University of Illinois Urbana-Champaign

Marco Morales^{||}

ITAM

University of Illinois Urbana-Champaign

Nancy M. Amato^{**}

University of Illinois Urbana-Champaign

ABSTRACT

Many tasks are intuitive for humans but difficult to encode algorithmically when utilizing a robot. Robotic systems often benefit from learning from expert demonstrations, wherein operators move the robot along trajectories. Often, using a physical robot to provide these demonstrations may be difficult or unsafe. Extended reality provides a natural setting for demonstrating trajectories while bypassing safety concerns or modifying existing solution trajectories. We propose the Robot Action Demonstration in Extended Reality (RADER) system for learning from demonstration approaches. We additionally present its application to a state-of-the-art approach and show comparable results between using a physical robot and our system.

Index Terms: Computing methodologies—Learning from demonstrations, Human-centered computing—Mixed / augmented reality, Human-centered computing—Virtual reality

1 INTRODUCTION

In the real world, there are many tasks that humans can accomplish more efficiently and reliably than robots such as cooking tasks like slicing, scooping, or mixing [1]. In these situations, robotic systems can benefit from human demonstrations. These demonstrations often consist of trajectories generated by the human via controller-based teleoperation or by directly dragging parts of the robot [5].

Providing these demonstrations can often be difficult or unsafe. Many real-world environments may be challenging to replicate in a lab setting. For example, tasks like slicing feature robots handling potentially dangerous tools like knives. We propose an *Extended reality* (XR) system to allow users to safely and effectively provide demonstrations in virtual reality (VR) or augmented reality (AR).

Prior work has explored AR interfaces for human demonstration collection. Leubbers et al. [4] present an interface that visualizes constraints the robot is subject to while performing its task, enabling the human to comply with these constraints while providing their demonstration. This system, however, relies on a physical robot, limiting its potential use cases. Lotsaris et al. [3] present an AR system that allows a user to position a virtual gripper in the environment. Here, the full robot trajectory is computed by a motion

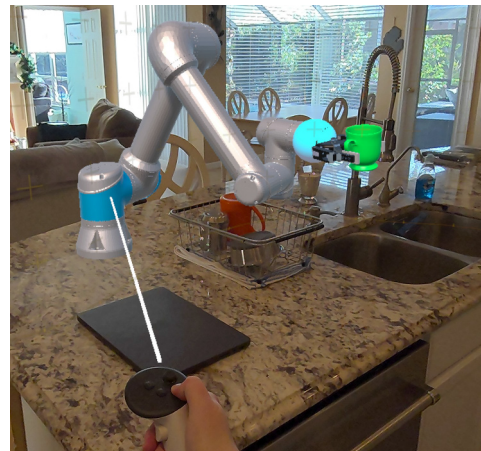


Figure 1: An example of a user interacting with a robotic manipulator to direct it away from a laptop while carrying a mug. The user can directly adjust each robot joint or drag the target sphere attached to the end effector to have the robot follow the user's motion.

planner rather than provided by the user. Our system aims to let a human provide complete robot trajectories using a virtual robot.

We evaluate our system using the Feature Expansive Reward Learning (FERL) method. [2]. This approach to inverse reinforcement learning recovers a reward function over a robot's state space by collecting human input in the form of *feature traces*, trajectories that move monotonically from more desirable to less desirable states. In real-world applications, collecting these feature traces on a physical robot may require the human to place the robot in an unsafe or undesirable configuration. We recognize that this approach could benefit from using a virtual robot to collect feature traces.

2 APPLICATIONS

We anticipate a wide range of applications for our system, including manufacturing and domestic scenarios. In industrial settings, digital twins allow users to simulate the performance and safety of a system and, thus to make informed design decisions, as well as train a robot before deployment. Our system may also enhance assistive robotic systems designed to support activities of daily living. Those with disabilities may struggle to manually provide feedback by pushing or dragging parts of a robot; however, using a controller to provide input may be possible. We expect our system to provide the most benefit for tasks requiring complex motions where human demonstrations can be especially valuable. In a kitchen environment, for example, this could include tasks such as retrieving items from a cluttered refrigerator or manipulating tools to prepare food.

*e-mail: imngui2@illinois.edu

[†]e-mail: cmcbeth2@illinois.edu

[‡]e-mail: gche3@illinois.edu

[§]e-mail: andres11@al.insper.edu.br

[¶]e-mail: lpsoares@insper.edu.br

^{||}e-mail: moralesa@illinois.edu

^{**}e-mail: namato@illinois.edu

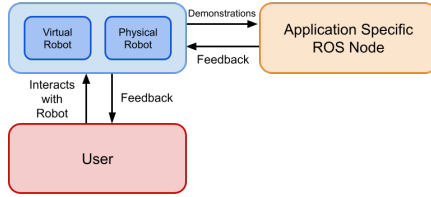


Figure 2: As the user interacts with the robot, collected trajectories are sent to application ROS nodes, which collect and process the data. Trajectories may be executed on a physical robot.

3 PROPOSED SYSTEM

Our system supports both VR and AR. In the AR setup, the system allows the user to choose where to place the virtual robot's base before interacting with it. We intend to support many virtual reality headsets and are using the OpenXR API¹ for VR and AR functionality, which is compatible with most headsets currently available. Additionally, we use the Unity Game Engine² to build our system due to its compatibility with a broad range of XR devices and wide availability of open-source packages. We support any manipulator defined with a URDF file, a standard format in robotics.

3.1 User Interface

In the virtual environment, the user can interact with the robot through their hand controller to grab the end effector or any other link (Fig. 1). The user can press a button to start trajectory recording. Haptic feedback in the handheld controllers can also be used to indicate when a collision is imminent by setting the intensity of the feedback proportional to the distance of a robot from the nearest obstacle. Inverse kinematics solutions are computed as the trajectory is constructed to recover the full robot configuration and accurately display the robot. Depending on the application, the system may be configured to provide feedback including, for example, how much of the robot's reachable workspace is covered by trajectory demonstrations. In industrial applications, feedback in a VR digital twin setup may include annotating provided trajectories with safety information such as distance from human workstations. In domestic settings, AR may be more beneficial, allowing the user to visualize a virtual robot performing tasks in the real home environment to better gauge performance before executing on a physical robot.

3.2 Communication Between Unity and ROS

We build the virtual environments in Unity. Unity offers integration with the Robot Operating System (ROS)³, enabling us to send information from our virtual reality application to ROS nodes we create. These nodes collect the human-demonstrated trajectories, which can be passed on to be used in downstream processes.

3.3 Downstream Processes

After demonstration collection, a ROS node stores the data. The data may be processed (e.g., smoothed) before being stored. Downstream processes can take these demonstrations as training data or execute them on a physical robot.

4 EVALUATION

We compare the results of learning from demonstration using FERL [2] using demonstrations provided on a physical UR5e robot and a virtual robot in RADER. We learn a reward function over the

¹OpenXR API: <https://www.khronos.org/openxr/>

²Unity Game Engine from Unity Technologies: <https://unity.com/>

³Unity ROS Connector: <https://github.com/Unity-Technologies/ROS-TCP-Connector/>

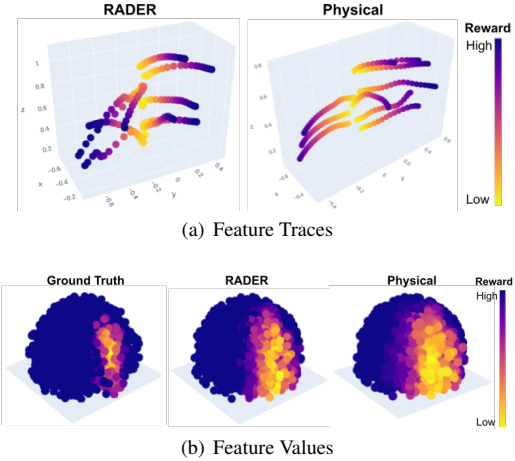


Figure 3: Feature traces and feature values for the laptop-avoidance scenario.

robot's state space that instructs the robot to avoid moving over a laptop on the table (Fig. 1). We show a comparison between the feature values learned by each method in Figure 3. The mean squared error between the feature values learned using our system and the physical robot averaged over 10,000 randomly sampled points in the state space are approximately 1.6 and 3.8 respectively. This indicates that our system can be used to generate demonstrations as informative as those given using a physical robot.

5 CONCLUSIONS AND FUTURE WORK

We present RADER⁴, our XR demonstration collection system. We intend this system to enable users to provide expert demonstrations to robotic systems without requiring access to a potentially costly physical robot. We anticipate that our system will provide benefits when collecting physical robot demonstrations would be unsafe or when users are collaborating on projects remotely. In the future, we intend to evaluate our system with an extensive user study.

ACKNOWLEDGMENTS

This work supported in part by the Illinois-Inspire Partnership at the University of Illinois and by Asociación Mexicana de Cultura AC.

REFERENCES

- [1] C. C. Beltran-Hernandez, N. Erbeti, and M. Hamaya. Sliceit! - a dual simulator framework for learning robot food slicing. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 4296–4302, 2024. doi: 10.1109/ICRA57147.2024.10611596 1
- [2] A. Bobu, M. Wiggert, C. Tomlin, and A. D. Dragan. Feature expansive reward learning: Rethinking human input. In *2021 16th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, pp. 216–224, 2021. 1, 2
- [3] K. Lotsaris, C. Gkournelos, N. Fousekis, N. Kousi, and S. Makris. AR based robot programming using teaching by demonstration techniques. *Procedia CIRP*, 97:459–463, 2021. 8th CIRP Conference of Assembly Technology and Systems. doi: 10.1016/j.procir.2020.09.186 1
- [4] M. B. Luebbbers, C. Brooks, M. J. Kim, D. Szafrir, and B. Hayes. Augmented reality interface for constrained learning from demonstration. In *Proceedings of the 2nd International Workshop on Virtual, Augmented and Mixed Reality for HRI (VAM-HRI)*, vol. 142, 2019. 1
- [5] H. Ravichandar, A. S. Polydoros, S. Chernova, and A. Billard. Recent advances in robot learning from demonstration. *Annual review of control, robotics, and autonomous systems*, 3:297–330, 2020. 1

⁴<https://github.com/parasollab/RADER/releases/tag/v1.0>